

What is a Transitive Dependency?

Formal Definition

A Transitive Dependency exists when:

$X \rightarrow Y$

$Y \rightarrow Z$

therefore

$X \rightarrow Z$

where:

- X is a Candidate Key (or Super Key)
- Y is a non-key attribute
- Z is another non-key attribute

Direct Dependency

A direct dependency means an attribute depends immediately on the Candidate Key.

Indirect Dependency

An indirect dependency passes through another attribute.

Example

`StudentID → DepartmentID`

`DepartmentID → DepartmentName`

Identification of Transitive Dependency

Follow these steps.

- **Step 1** Find the Candidate Key.
- **Step 2** Find all non-key attributes.
- **Step 3** Check whether one non-key attribute determines another non-key attribute.

If yes,

Transitive Dependency exists.

Why is Transitive Dependency Bad?

Suppose:

EmpID	DepartmentID	Manager
1	ECE	Dr. Sharma
2	ECE	Dr. Sharma
3	ECE	Dr. Sharma

Manager name repeats many times.

Problems:

- Redundancy
- Update anomaly
- Storage waste
- Data inconsistency

Removing Transitive Dependency (3NF)

Original Table

EmpID	EmployeeName	DepartmentID	Manager
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Split into:

Employee

EmpID	EmployeeName	DepartmentID
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Department

DepartmentID	Manager
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Now

EmpID → DepartmentID
DepartmentID → Manager

No redundant manager data exists.
This satisfies Third Normal Form (3NF).

What is a Multivalued Dependency?

Definition

A Multivalued Dependency (MVD) exists when one attribute determines multiple independent values of another attribute.

Independent Multivalued Attributes

Consider

- Student
- Programming Language
- Spoken Language

Programming Languages

- C++
- Python

Spoken Languages

- English
- Hindi

Learning Python does not affect which spoken languages Aadya knows.

Therefore,
the two multivalued attributes are independent.

Why MVD Causes Redundancy

Original

Student	Skill	Language
Aadya	C++	English
Aadya	C++	Hindi
Aadya	Python	English
Aadya	Python	Hindi

Rows required

Skills × Languages

$$2 \times 2 = 4$$

If Aadya later learns Java,
rows become

$$3 \times 2 = 6$$

Growth is multiplicative.

Removing MVD (4NF)

Split into

StudentSkill

Student	Skill
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StudentLanguage

Student	Language
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Now no unnecessary combinations exist.
This satisfies Fourth Normal Form (4NF).

Complete Comparison

Feature	Partial Dependency	Transitive Dependency	Multivalued Dependency
Depends on	Part of a Composite Key	Another non-key attribute	Multiple independent values
Violates	2NF	3NF	4NF
Requires Composite Key?	✔ Yes	✗ No	✗ No
Main Problem	Redundancy due to partial determination	Redundancy due to indirect determination	Redundancy due to independent multivalued facts
Example	StudentID → StudentName where key is (StudentID, CourseID)	DepartmentID → HOD	Student ↔ Skill, Student ↔ SpokenLanguage